

Answer **all** questions.

1. The diagram shows the 7 regions of the electromagnetic (em) spectrum in order.



gamma rays	A	B	visible light	infra-red	microwaves	radio waves
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(a) Use **only** the words given in the box below to answer the following questions.

cosmic rays	X-rays	ultrasound	water	ultraviolet
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- (i) Name the region **labelled A**.

[1]
- (ii) Name the region **labelled B**.

[1]
- (b) (i) Complete the following sentence by underlining the correct word in the bracket. [1]

The arrow shown on the electromagnetic spectrum above represents increasing (**amplitude** / **wavelength** / **frequency**).

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only

(ii) Waves in the electromagnetic spectrum all travel at the same speed in a vacuum.

Tick (✓) the **three** correct statements that describe other properties of em waves. [3]

☐

All transverse.

☐

All ionising.

☐

All transfer energy.

☐

All longitudinal.

☐

All travel through space.

☐

All given out by radioactive materials.

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